

LOKESH REDDY THIPPARTHI

AI / ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer and Data Scientist with 4+ years of progressive experience building and maintaining ML systems across the full lifecycle. Started by contributing to data pipelines and model experiments, and have since taken ownership of production deployments, monitoring, and MLOps workflows. Experienced with **MLOps**, **Generative AI (RAG)**, **NLP**, and cloud platforms including **AWS** and **GCP**. Comfortable working in collaborative teams and iterating quickly on real-world data problems.

PROFESSIONAL EXPERIENCE

Machine Learning Analyst | SPR Software Systems

Jan 2024 – Present

The colony, TX

- Models in production had no consistent way to catch degradation early — **designed and implemented a drift detection pipeline** using AWS CloudWatch and SageMaker Model Monitor that surfaced issues before they impacted downstream reports, cutting manual review cycles by around **30%**.
- Model releases were slow and involved too many manual steps — **set up CI/CD pipelines with MLflow and Jenkins** to standardize versioning and deployments, making releases more reliable and rollbacks straightforward.
- Training large models was a bottleneck for the team's iteration speed — **moved training workloads to distributed TensorFlow on cloud infrastructure**, reducing training time by roughly **40%** and enabling faster experimentation cycles.
- Inference latency was affecting time-sensitive financial pipelines — **optimized model serving using Kubernetes and ONNX runtimes**, bringing average latency down by around **60%** on critical workflows.
- Integrated **Model Context Protocol (MCP)** into an internal AI agent to improve how it selected and used tools — made the system more predictable and easier for non-ML teammates to interact with.
- With models informing financial decisions, compliance was a consistent requirement — **implemented Responsible AI (RAI) checks** for bias and fairness monitoring, providing documented evidence of model behavior during audits.

Data Scientist | Krithi Cyber Soft PVT LTD

Jun 2020 – Jul 2022

Hyderabad, India

- The existing fraud detection model had a high false positive rate causing alert fatigue — **re-examined the feature set, introduced new derived features, and ran hyperparameter tuning with Ray Tune**, improving the model's F1-score from **0.82 to 0.94** across several iterations.
- Analysts were spending too much time manually preparing data before each modeling run — **contributed to building PySpark scripts on Databricks** that automated key steps in the data preparation workflow, noticeably reducing prep time for the team.
- A recurring pain point was data arriving in inconsistent formats from different sources — **wrote validation scripts and transformation logic in Python and SQL** that standardized inputs before they reached the modeling layer, reducing pipeline failures.
- Gained hands-on experience with **PyTorch and Hugging Face Transformers** by building a document classification model that categorized incoming reports automatically, reducing items being manually triaged.
- Supported data delivery workflows connecting **Snowflake and Redshift** to downstream model pipelines — added schema validation checks that caught data issues early and improved team confidence in production data.

CORE COMPETENCIES

ML & MLOps	TensorFlow, PyTorch, scikit-learn, MLflow, Distributed Training, Model Monitoring, A/B Testing, ONNX, CUDA
Languages	Python, SQL, Java, PySpark, NumPy, Pandas, JAX, FastAPI
Generative AI	LangChain, RAG Pipelines, OpenAI API, Hugging Face Transformers, NLP, Computer Vision, Transfer Learning, MCP
Cloud & DevOps	AWS SageMaker, Google Vertex AI, Kubernetes, Docker, Jenkins, Apache Airflow, Kafka, EC2, S3, Lambda
Data Engineering	Databricks, Snowflake, ETL/ELT Pipelines, FAISS, Pinecone, Redshift, Delta Lake, Spark SQL

EDUCATION

Master's in Information Sciences and Systems – Trine University, Angola, Indiana

Bachelor's in Computer Science Engineering – Jawaharlal Nehru Technological University, Hyderabad, India

CERTIFICATIONS

- Oracle Cloud Infrastructure Certified Generative AI Professional - [Badge](#)
- Oracle Cloud Infrastructure Certified AI Foundations Associate - [Badge](#)
- Databricks Certified Generative AI Engineer Associate - [Badge](#)